SILFARADILA MEI FIZAHIDAH

COMPUTER VISION ENGINEER | AI ENGINEER

Yogyakarta, Indonesia • silfaradilamei@gmail.com • +62 898-5358-489 • www.linkedin.com/in/silfaradila/

SUMMARY

AI Engineer with hands-on experience developing computer vision systems, and LLM-powered applications. Experienced in end-to-end model development, RAG pipelines, LLM evaluation, backend/API engineering, and deploying computer vision solutions in operational environments. Combines practical machine learning expertise with algorithmic problem-solving and system integration.

EXPERIENCE

BIOLOGIC TECHNIK PTE. LTD. **Singapore - Remote**

Computer Vision Engineer **July 2025 – Present**

As a Computer Vision Engineer, I am responsible for developing and improving computer vision solutions, from data collection and preprocessing to model training, evaluation, and deployment. I worked on object detection, segmentation, visual recognition, and anomaly detection. I also developed backend integrations, collaborated with infrastructure and IoT teams, supported system deployment, monitored model performance, analyzed errors, and prepared technical reports to guide further improvements.

Project Experience

Smart Inventory Management System (SIMS)

Developed and improved a computer vision-based inventory monitoring and counting system deployed in a hospital inventory environment.

- Trained and evaluated custom YOLOv8 segmentation models to identify visible surfaces of stacked boxes; developed geometry-based counting logic using bounding-box coordinates, adjacency, and spatial relationships to estimate hidden boxes.
- Analyzed daily counting discrepancies across model, counting-logic, and environmental failures; improved the system through targeted data collection, augmentation, and counting-logic refinements.
- Reported peak average daily operational inventory-counting accuracy of 98.6%; prepared monthly reports covering accuracy, PAR performance, failure categories, and improvements.
- Proposed a centralized inference architecture to reduce duplicate YOLO model loading and instability on Raspberry Pi devices; personally developed the FastAPI inference server, integrated YOLO execution, and configured Redis-based task processing.
- Collaborated with infrastructure engineers on AWS deployment and integration with existing MQTT communication. The deployed setup used seven cameras across seven shelves and four Raspberry Pi 4 devices.

CSSD Surgical Instrument Identification & Abnormality Detection

Handled end-to-end development for surgical instrument identification and surface-abnormality investigation, including the FastAPI backend, PostgreSQL integration, and AWS EC2 staging deployment.

- Built an identification pipeline combining YOLO detection, image preprocessing, Segment Anything Model (SAM) mask refinement, geometric length estimation using PCA, and fixed-camera pixel-to-centimeter calibration.
- Integrated DINOv2 embeddings, cosine similarity, FAISS, and PostgreSQL/pgvector to retrieve candidate instruments, with physical-length metadata used for candidate filtering.
- Evaluated 11 instrument types, with approximately 10 correctly identified at top-1 on the small test set.
- Explored blood-stain, rust, and damage detection using HSV processing, PatchCore, and a custom patch-based approach with HSV/grayscale/LAB features and Isolation Forest/SVM; these methods remained under development.

LLM Market Sensing & Website AI Assistant

Contributed to an LLM-powered market intelligence platform, focusing on automated market sensing and an independently implemented website RAG assistant.

- Built an n8n-based daily news collection workflow, using OpenAI to extract structured JSON from Singapore-related news and filter relevant stories while deduplicating multiple articles about the same event.
- Stored filtered news in PostgreSQL and implemented scheduled signal processing that combined external news with internal SKU and supplier information. Proprietary signal-calculation rules are excluded.
- Independently implemented an end-to-end RAG assistant using FastAPI, AnythingLLM API, and ChromaDB, supporting user-selectable OpenAI and Claude models and queries about forecast results and market-sensing information.
- Integrated JSON, TXT, and CSV platform data with dynamically extracted webpage content: JavaScript captured the current page, the content was chunked, and embeddings were generated at runtime to support page-aware questions.
- Built a custom LLM-powered SQL agent using FastAPI and PostgreSQL to answer natural-language questions about live forecast data. Restricted its shared database account to read-only access and relevant forecast tables.
- Designed a RAG evaluation workflow combining manual fact-checking against reference facts and cited documents with DeepEval (faithfulness, completeness, instruction following) and TruLens (relevancy).

Clinical Laboratory Staff Scheduling & Optimization System

Developed a workforce scheduling and optimization system for clinical laboratory operations using Python, FastAPI, PostgreSQL, and constraint-programming techniques.

- Built a Greedy Algorithm scheduling engine with multi-factor heuristic scoring to assign staff to shifts based on skills, qualifications, availability, shift preferences, and working-hour constraints.
- Implemented an optional Google OR-Tools CP-SAT optimization engine using Boolean decision variables, hard constraints, and objective maximization to optimize staffing assignments.
- Developed automated urgent-leave handling to identify affected shifts and assign qualified replacements while validating availability and weekly working-hour limits.
- Designed a seven-metric roster evaluation framework covering staff utilization, skill matching, shift coverage, workload fairness (Gini coefficient), shift preferences, supervisor allocation, and cost efficiency.
- Built workload analytics and optimization recommendations to identify understaffing, skill gaps, workload imbalances, and inefficient resource allocation; prepared demonstration scenarios and technical documentation comparing heuristic and constraint-programming approaches.

CV. KARYA HIDUP SENTOSA (QUICK TRACTOR) **Yogyakarta, Indonesia**

Junior Machine Learning Engineer **June 2024 – June 2025**

I progressed from an intern to a full-time Junior Machine Learning Engineer in June 2024, maintained and monitored machine learning models and services using Prometheus and Grafana, and researched and evaluated emerging AI technologies to identify innovative solutions for potential company implementation.

Project Experience

WorkGuard: Violation Detection in Workplace

Developed a real-time workplace violation detection system using YOLOv8, leveraging advanced video analytics to identify infractions like unauthorized phone use, unsafe loads, and restricted area violations.

- Designed and architected the complete system infrastructure, including data pipeline, model training workflow, and deployment architecture for scalable real-time processing.
- Collected, annotated, and processed a comprehensive dataset with 10K+ images across multiple violation categories, implementing class-balanced data augmentation techniques (rotation, brightness adjustment, noise injection) to ensure equal representation and prevent model bias.
- Optimized YOLOv8 performance through Optuna-driven hyperparameter tuning and CometML experiment tracking for efficient model versioning, achieving 92% mAP score and 45ms inference time.
- Built scalable backend architecture using FastAPI and Celery to handle concurrent CCTV streaming from 15+ cameras simultaneously, implementing asynchronous task queues for real-time video processing without performance degradation.

- Deployed the detection model via RESTful API endpoints for real-time monitoring and implemented an automated email alert system using SMTP integration for immediate violation notification to supervisors.
- Integrated automated PDF report generation with violation analytics dashboard for comprehensive tracking and compliance reporting, resulting in a 25% reduction in workplace violations and enhanced safety compliance across the facility.

Face Attendance System

Leveraging facial recognition and anti-spoofing techniques, this system provides secure and accurate employee attendance tracking.

- Collect employee photos for face data enrollment and perform data preprocessing for anti-spoofing datasets.
- Design and develop a scalable classification model for anti-spoofing using TensorFlow.
- Deployed the anti-spoofing model via TensorFlow Serving for production-ready implementation.
- Achieved face recognition accuracy of 87% with Dlib and 92% with FaceNet.
- Developed a Flask REST API to provide seamless access for front-end applications.
- Optimized model inference for resource efficiency, reducing latency and recognition time to 0.5 seconds per image.

CV. KARYA HIDUP SENTOSA (QUICK TRACTOR) **Yogyakarta, Indonesia**

Internship **June 2023 - June 2024**

- Delivered practical AI solutions, including computer vision and machine learning models, that directly improved operational efficiency and safety for the company.
- Implemented data engineering and analytics systems to transform raw data into actionable insights, driving business growth.
- Built CBIR system using ResNet and VGG19 models to generate feature vectors for visual product search, improving search accuracy and user experience for 'TokoQuick' e-commerce platform.
- Developed YOLOv5-based automated counting system for materials detection, achieving 97% accuracy through hyperparameter optimization and enabling efficient inventory management.

EDUCATION

SMK N 2 KLATEN **Klaten, Central Java, Indonesia**

High School; Major in Informatics Systems Engineering, Applications, and Networking **2020 - 2024**

SKILLS

Computer Vision & ML	Python, PyTorch, TensorFlow/Keras, YOLOv5/YOLOv8, SAM, DINOv2, ResNet, VGG19, Scikit-learn, Isolation Forest, SVM
LLM & Applied AI	OpenAI API, Anthropic Claude, RAG, AnythingLLM, ChromaDB, n8n, DeepEval, TruLens, Prompt Engineering, Custom SQL Agents
Backend & Data	FastAPI, Flask, PostgreSQL, pgvector, FAISS, SQLAlchemy, Redis, MQTT, Python asyncio, REST APIs
Infrastructure	AWS EC2, AWS S3, Docker, Linux, Uvicorn, Raspberry Pi, Prometheus, Grafana

LANGUAGES

Bahasa Indonesia - Native Proficiency

English - B2 Level of Proficiency